

TubulePy

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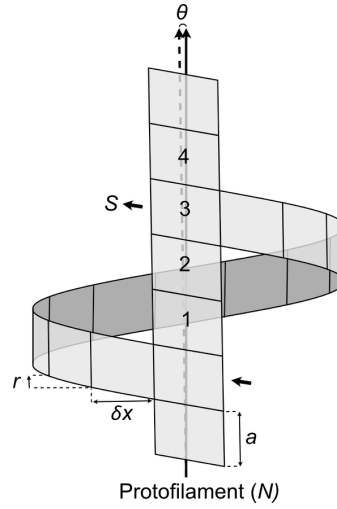
Introduction

TubulePy has been designed to compute the protofilament skew angle (θ) of microtubules with different protofilament numbers (N) and monomer helical-start numbers (S), taking into account variations in their main lattice parameters:

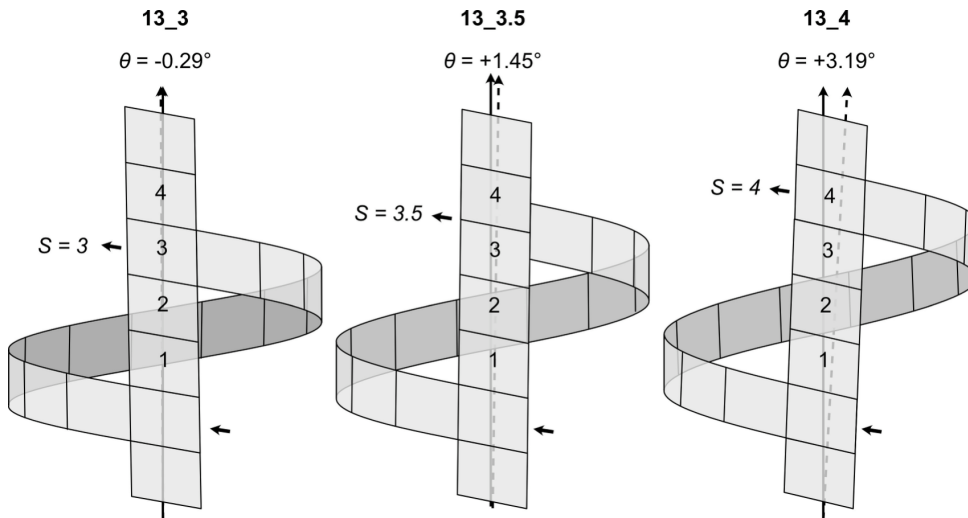
- the subunit repeat (a) along protofilaments, reflecting the compaction state of tubulin,
- the rise (r) of the subunits along the left-handed S -start helices,
- the separation between protofilaments (δx).

θ is given by (Chrétien & Fuller, 2000):

$$\theta = \tan^{-1} \left(\frac{1}{\delta x} * \left(\frac{S*a}{N} - r \right) \right) \quad (\text{eq.1})$$



- Microtubules built from **A-** and/or **B-type** lateral interactions will have integer values of S , (e.g., 2, 3 or 4), while those incorporating a **C-** or a **D-type** lateral interaction between two protofilaments will have fractional ones (e.g., 2.5 or 3.5) (Guyomar *et al.*, in preparation).



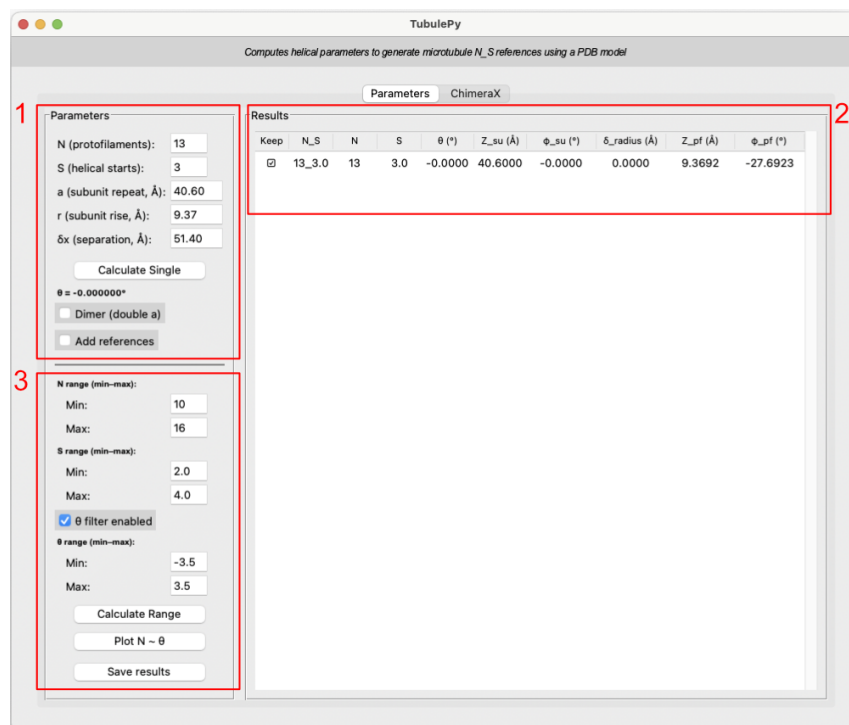
Theoretical protofilament skew angles corresponding to GDP-BeF₃⁻ microtubules, with a tubulin spacing (a) of 40.6 Å, and a rise (r) of 9.63 Å.

Installation

- Download TubulePy.py and the PDB files at: <https://github.com/dchretien35/TubulePy>
- Place the Alpha.pdb, Alpha_Beta.pdb and Alpha_Beta_kin.pdb files in a separate folder named PDB_model.
- It is recommended to create a TubulePy environment:
 - Install [miniconda3](#) (if not yet present in your system):
 - Open a terminal, and at the command line run:
`conda create --name tubulepy-env python=3.11 (or your Python version)`
 - Then enter: `conda activate tubulepy-env`
 - To quit the environment, use: `conda deactivate`
- [Python](#): 3.9+ versions recommended (runs fine on 3.11/3.13).
- Needs Tkinter available with the Python build (standard on most macOS/Windows installers; on some Linux distros: run `'install python3-tk'`).
- External app: Latest version of [UCSF ChimeraX](#) installed and its ChimeraX binary accessible (path is user-selectable in the app).
- Packages for full feature set (projections and gallery): numpy, mrcfile, Pillow.
- Optional packages: matplotlib (needed for the $N \sim \theta$ plot and as a fallback saver if Pillow is missing).
- One-line install (inside your desired env): run `'pip install numpy mrcfile Pillow matplotlib'` in the terminal before running TubulePy.

Procedure

- Once in your environment, type at the terminal: `python TubulePy.py`



- **TubulePy** is composed of two tabs (**Parameters** and **ChimeraX**). The first tab allows calculation of protofilament skew angles and helical parameters of microtubule N_S configurations. The second tab allows computation of PDB-based microtubule models, MRC derived 3D models, as well as projected densities using ChimeraX.

Parameters

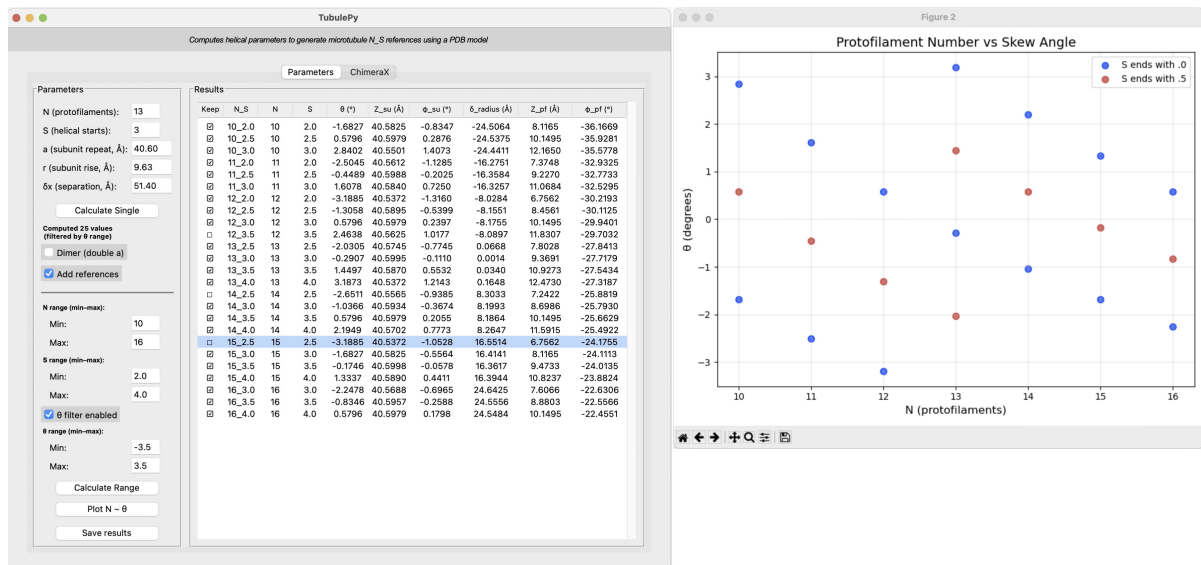
Panel (1): allows calculations for a single microtubule with the indicated parameters.

- '**Calculate single**' writes the protofilament skew angle (θ) and the helical parameters (see annex 2) for this N_S configuration in the Results window.
- Results can be saved in a text file using '**Save results**' at the bottom of **panel (3)**.
- Several microtubules with different N_S configurations can be computed by checking '**Add references**'.
- Helical parameters for dimers can be computed by selecting '**Dimers (double α)**'.

Panel (2): Results window. A checkbox is provided to select (or deselect) N_S configurations before saving.

Panel (3): allows calculations for a range of N_S configurations with the lattice parameters indicated in Panel (1).

- **N range (min-max):** Protofilament number range.
- **S-range (min-max):** monomer helical-start number range.
- **θ filter enabled:** Checked by default. Avoids calculation of excessive protofilament skew angles outside the 'Min and Max' values to reflect the limited torsional flexibility of the microtubule shaft.
- **Calculate Range:** Calculates the protofilament skew angles and helical parameters of the different N_S configurations and list them in the result window.
- **Plot $N \sim \theta$:** Plots the protofilament skew angles of the different N_S configurations. Microtubules with integer values of S are indicated in blue, and fractional ones in red.

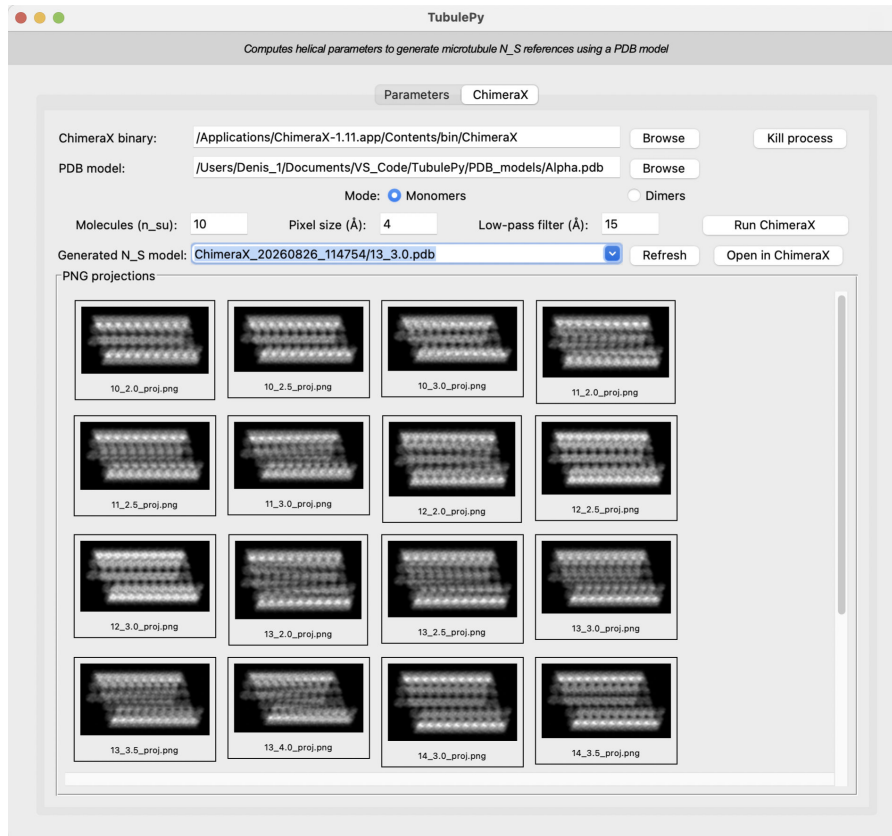


N_S configurations with their corresponding θ plot. Some configurations have been unchecked.

- **Save results:** Saves results in a Ref_date_time.txt file, which can be used to generate references for 2D classification for instance.

Most users will stop here since they will use their own methods to generate references from this table. Command lines that can be used in ChimeraX are provided in annexes 3 and 4. Saving this file is not necessary to go to the next tab.

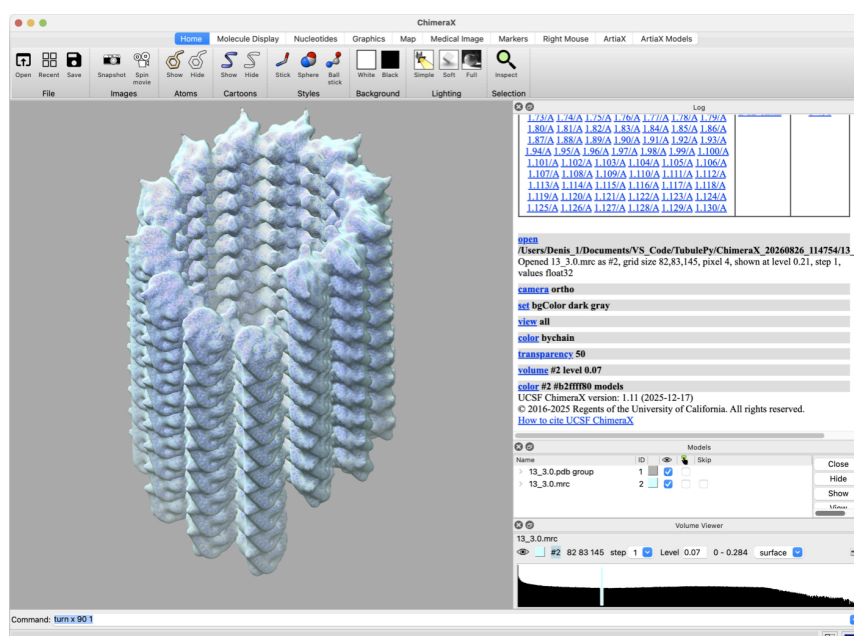
ChimeraX



This tab is used to generate PDB models that are transformed into densities (in [.mrc format](#)). Projections of these 3D density models are calculated and saved in [.png format](#).

- **ChimeraX Binary:** Indicates the path to the ChimeraX application. In macOS, this is the default path: /Applications/ChimeraX-1.11.app/Contents/bin/ChimeraX. The path is recorded once defined in this menu.
- **PDB model:** Path to the PDB_models subfolder. This folder contains three files derived from the [3J8Y PDB model](#) (Shang et al., 2014): the full dimer with kinesin (Alpha_Beta_kin.pdb), the dimer without kinesin (Alpha_Beta.pdb), and the alpha monomer (Alpha.pdb). Other models can be imported and placed in this folder, but they will have to be fitted to one of these original models and saved with respect to their coordinates.
- **Mode:** Used to compute N_S configurations from **Monomers** (e.g., Alpha.pdb) or **Dimers** (e.g., Alpha_Beta.pdb or Alpha_Beta_kin.pdb).
- **Molecules (n_su):** Number of 'molecules' that will be modelled in a single protofilament. This will determine the length of the microtubule that will be generated. Note that for the same length, a dimer version will require two-times less molecules.

- **Pixel size (Å):** Pixel size of the generated N_S.mrc files. In a 2D classification approach, this is the pixel size of the cryo-EM images.
- **Low pass filter:** this parameter allows filtering the N_S.mrc files to the desired resolution. A too high level of details might bias a 2D classification approach.
- **Run ChimeraX:** runs ChimeraX at the command line without its graphical interface. A subfolder is created named ChimeraX_date_time that contains all the PDB, MRC and PNG files.
- **Kill process:** can be used to stop the computation of the N_S configurations. A message indicating 'Process running...' is indicated next to it until it is completed.
- **Generated N_S model:** Allows selection of a N_S.pdb model that can be opened using the 'Open in ChimeraX' button. The list of models present in the ChimeraX_date_time subfolder can be refreshed.
- **Open in ChimeraX:** Opens the selected .pdb model and .mrc file in ChimeraX for inspection.



- **PNG projections:** this window displays all the projections that have been generated. One interest of TubulePy is to build long microtubules and compare their projections and FFTs with those of experimental images (after contrast inversion). This allows characterization of unknown or difficult to analyze configurations, e.g., those incorporating C- or D-type lateral interactions.

Annex 1: Determination of microtubule lattice parameters

A crucial step before using TubulePy is to retrieve basic lattice parameters for the specimen of interest. This can be done with [TubuleJ](#) that allows analysis of cryo-EM images of microtubules and of their Fourier transforms.

- In practice, the average subunit repeat (a) must be known or determined using TubuleJ from the position of the 4 nm^{-1} layer line in the microtubule FFTs, eventually corrected

by the protofilament skew angle (correction taken into account in TubuleJ). These measurements can be performed on a few tens of cryo-EM images of microtubules before running TubulePy.

- Once determined, the second important parameter is the subunit rise (r) along the left-handed helices, which is given for each microtubule analyzed by:

$$r = \frac{S * a}{N} - \delta x * \tan(\theta) \quad \text{eq. (2)}$$

- The separation between protofilaments (δx) can be let to its default value of 51.4 Å (Chrétien et al., 1998).

Annex 2: Determination of microtubule helical parameters

To build helices, we use six formulas, which are the six last columns in the Results window of TubulePy. The rational is to first build a protofilament (eqs. 3-5), and then the full microtubule from this protofilament (eqs. 6-7).

- The rise of tubulin subunits (Å) along the Z-axis:

$$\Delta z_{su} = \cos(\theta) * a \quad \text{eq. (3)}$$

Double this value in case of dimers.

- The rotation angle (°) of the subunits with respect to the Z-axis:

$$\Delta \varphi_{su} = \frac{\sin(2 * \theta) * a * 180}{N * \delta x} \quad \text{eq. (4)}$$

Double this value in case of dimers.

- The radial shift (Å) of the helices with respect to the 13_3 position used as a reference:

$$\delta c = \frac{\delta x}{2 * \pi} * \left(\frac{N}{\cos(\theta)} - 13 \right) \quad \text{eq. (5)}$$

- The rise of the protofilaments (Å) with respect to the Z-axis:

$$\Delta z_{pf} = \sin(\theta) * \delta x + \cos(\theta) * r \quad \text{eq. (6)}$$

- The rotation angle (°) of the protofilaments with respect to the Z-axis.

$$\Delta \varphi_{pf} = \frac{S * \Delta \varphi_{su} - 360}{N} \quad \text{eq. (7)}$$

Annex 3: ChimeraX commands implemented to generate the PDB models

Adapted from an original tutorial of Ken Downing on ChimeraX web site, but which is not available anymore. Refer to the names provided in the Ref_date_time.txt file.

The following commands have been implemented in TubulePy:

- Two copies of a PDB model are opened in ChimeraX (e.g., Alpha.pdb), which will have IDs #1 and #2.

- Subunit (#2) is tilted by θ

```
turn 0,1,0 Theta_deg models #2 coordinateSystem #1
```

- A protofilament is built by moving the subunit by ΔZ_{su} and turning it by $\Delta\phi_{su}$

```
sym #2 h,r_su_A,Phi_su_deg,n,-n/2 coord #1 center 0,Delta_C_A,0 copies true
```

where 'n' is the number of 'molecules' (n_{su} in ChimeraX panel).

- Helical symmetry is applied to this protofilament using ΔZ_{pf} and $\Delta\phi_{pf}$, making N-1 copies to complete the microtubule wall:

```
sym #3 h,r_pf_A,Phi_pf_deg,N coord #1 center 0,Delta_C_A,0 copies true
```

- The model is centered:

```
move 0,1,0 -Delta_C_A coord #4 models #4
```

Annex 4: ChimeraX commands implemented to generate MRC files

Chimera has a '[molmap](#)' function that transforms .pdb files into .mrc files. However, it cannot be used directly due to incompatibilities in coordinate systems. The following workaround works fine:

- Create an empty volume at the desired pixel size:

```
volume new gridSpacing {p_size} (4 Å by default in TubulePy).
```

- Match this volume to the .pdb model:

```
volume cover #5 atomBox #4
```

- Run the molmap command:

```
molmap #4 {lp_filter} onGrid #6
```

where 'lp_filter' is the filter applied to the .mrc file (15 Å by default in TubulePy).

References

- Chrétien, D., Flyvbjerg, H., & Fuller, S. D. (1998). Limited flexibility of the inter-protofilament bonds in microtubules assembled from pure tubulin. *European Biophysics Journal: EBJ*, 27(5), 490-500. <https://doi.org/10.1007/s002490050159>
- Chrétien, D., & Fuller, S. D. (2000). Microtubules switch occasionally into unfavorable configurations during elongation. *Journal of Molecular Biology*, 298(4), 663-676. <https://doi.org/10.1006/jmbi.2000.3696>
- Diaz, R., Rice, W. J., & Stokes, D. L. (2010). Fourier-Bessel reconstruction of helical assemblies. *Methods in Enzymology*, 482, 131-165. [https://doi.org/10.1016/S0076-6879\(10\)82005-1](https://doi.org/10.1016/S0076-6879(10)82005-1)
- Shang, Z., Zhou, K., Xu, C., Csencsits, R., Cochran, J. C., & Sindelar, C. V. (2014). High-resolution structures of kinesin on microtubules provide a basis for nucleotide-gated force-generation. *eLife*, 3, e04686. <https://doi.org/10.7554/eLife.04686>